

Alexis Buzzell

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Experience

Postdoctoral Scholar , Physics Education Research, Auburn University	Jan 2026 - Present
Graduate Student Researcher , Physics Education Research, University of Utah	Aug 2022 - Dec 2025

Education

University of Utah , PhD in Physics	May 2026
• Advisor: Ramón S. Barthelemy	
Worcester Polytechnic Institute (WPI) , MS in Mechanical Engineering	May 2020
Worcester Polytechnic Institute (WPI) , BS in Physics	May 2019
• Summa Cum Laude	

Publications

Published

- **Buzzell, A., Atherton, T., & Barthelemy, R.** (2025). Modern physics: Understanding the content taught in the U.S., *Physical Review Physics Education Research Focused Collection in Investigating and Improving Quantum Education through Research*, 21(1), 010139, <https://doi.org/10.1103/PhysRevPhysEducRes.21.010139>.
- **Buzzell, A., Atherton, T., & Barthelemy, R.** (2025). Quantum curriculum in the US: Quantifying the instructional time, content taught, and paradigms used, *Physical Review Physics Education Research Focused Collection in Investigating and Improving Quantum Education through Research*, 21(1), 010102, <https://doi.org/10.1103/PhysRevPhysEducRes.21.010102>.

Peer Reviewed Conference Proceedings

- **Buzzell, A., Atherton, T., & Barthelemy, R.** (2025). Assessing a combined human coding and natural language processing method for qualitative analysis in physics education research. *Physics Education Research Conference Proceedings*, <https://doi.org/10.1119/perc.2025.pr.Buzzell>.
- **Buzzell, A., & Barthelemy, R.** (2024). Certain bodies in uncertain fields: Thinking about gender through queer theory & quantum mechanics. *Physics Education Research Conference Proceedings*, <https://doi.org/10.1119/perc.2024.pr.Buzzell>.

In-Review

- **Buzzell, A., Atherton, T., & Barthelemy, R.** (2026). Characterization of the Graduate Level Quantum Curriculum within US Physics Doctoral Programs and Theoretical Frameworks of US Quantum Curriculum. Pre-print available at: <https://doi.org/10.21203/rs.3.rs-7745493/v1>.
- **Buzzell, A., Atherton, T., & Barthelemy, R.** (2026). *Survey on Faculty Perspective of the US Quantum Curriculum*.
- **Buzzell, A., Belcher, J., Ruben, R. Flint, P., & Burkholder, E.** (2026). How a physics orientation course impacted students' motivations to study physics.

Submitted

- **Buzzell, A., Burkholder, E., & Basir, S.** (2026). "We are treated as if we don't deserve to be here": First-year graduate students' accounts of their struggles in a physics doctoral program. *Physics Education Research Conference Proceedings*.

In Preparation

- **Buzzell, A., Atherton, T., & Barthelemy, R.** (2026). *Mixed methods approach for curriculum characterization and adequacy*.

- **Buzzell, A., Barthelemy, R., & Atherton, T. (2026).** *Use of natural language processes as unique method for validation in qualitative education studies.*

Awards

Physics Education Research Leadership and Organizing Council (PERLOC)	Jun 2025
Domestic Travel Grant, \$500	
Outstanding Graduate Teaching Assistant, Department of Physics & Astronomy, University of Utah, \$1000	Apr 2025
APS Group on PER (GPER) Journal Publication Fee Mini-Grant Award, \$500	Dec 2024
PERLOC Domestic Travel Grant, \$634	Apr 2024
GPER Conference Support Mini-Grant, \$1,000	Dec 2023
Swigart Fellowship, University of Utah	May 2023 - Aug 2023
Clare Booth Luce Research Scholar, WPI, \$6,000	Oct 2018 - May 2019
Summer Undergraduate Research Fellowship (SURF), WPI, \$5,000	Jun - Aug 2018
Nuclear Regulatory Commission (NRC) Scholarship, WPI, \$10,000	Jan - May 2018

Talks

Contributed

- **Buzzell, A., Atherton, T., & Barthelemy, R. (2026, May).** *Landscape of the U.S. Quantum Curriculum & Faculty Insights on Essential Knowledge* [Contributed talk]. The Southern Physics and Astronomy Graduate Student Symposium, Auburn, AL, USA.
- **Buzzell, A., Atherton, T., & Barthelemy, R. (2026, March).** *Mapping the US Graduate Quantum Curriculum* [Contributed talk]. American Physical Society Global Summit Meeting, Denver, CO, USA.
- **Buzzell, A., Atherton, T., & Barthelemy, R. (2026, March).** *Faculty Perspectives on Quantum Concepts in Graduate Physics Education* [Poster presentation]. American Physical Society Global Summit Meeting, Denver, CO, USA.
- **Buzzell, A., Atherton, T., & Barthelemy, R. (2025, August).** *Assessing a combined human coding and natural language processing method for qualitative analysis in physics education research* [Poster presentation]. Physics Education Research Conference Summer Meeting, Washington, DC, USA.
- **Buzzell, A., Atherton, T., & Barthelemy, R. (2025, March).** *Characterization of US institution's graduate quantum mechanics curriculum* [Contributed talk]. American Physical Society Global Summit Meeting, Anaheim, CA, USA.
- **Buzzell, A., Atherton, T., & Barthelemy, R. (2025, March).** *Characterization of the four-year undergraduate quantum curriculum across US institutions* [Poster presentation]. American Physical Society Global Summit Meeting, Anaheim, CA, USA.
- **Barthelemy, R., Buzzell, A., & Atherton, T. (2025, March).** *Characterization of the four-year undergraduate quantum curriculum across US institutions* [Contributed talk]. American Physical Society Global Summit Meeting, Anaheim, CA, USA.
- **Buzzell, A., & Barthelemy, R. (2024, July).** *Certain bodies in uncertain fields: Thinking about gender through queer theory & quantum mechanics* [Poster presentation]. Physics Education Research Conference Summer Meeting, Boston, MA, USA.
- **Buzzell, A., Barthelemy, R., & Atherton, T. (2024, July).** *Quantum curriculum in the US: Quantifying the instructional time, content taught, and paradigms used* [Contributed talk]. American Association of Physics Teachers Summer Meeting, Boston, MA, USA.
- **Buzzell, A., Barthelemy, R., Atherton, T., & Gerton, J. (2024, April).** *Modern physics: Understanding the content taught in the US* [Contributed talk]. American Physical Society April Meeting, Sacramento, CA, USA.

Teaching

Mentor Teaching Assistant, University of Utah	May 2025 - Jan 2026
• Developed and implemented course materials and curriculum for an online, asynchronous orientation for new departmental teaching assistants • Developed materials and curriculum for and lead the in person 2025 incoming graduate student orientation • Conducted observations of teaching assistants to provide feedback	

- Developed and lead monthly workshops for teaching assistants with the objective of furthering their teaching skills and professional development
Teaching Assistant, University of Utah Jan - May 2025
- Held regular office hours for Algebra-based Physics I course via Zoom
- Recorded problem solving tutorials for asynchronous online course
- Graded projects and exams
Teaching Assistant, University of Utah Aug - Dec 2024
- Held regular office hours for first-semester graduate-level Quantum Mechanics course
- Graded homework and exams
- Created solutions and grading rubrics for homework assignments
Teaching Assistant, University of Utah Jan - Apr 2024
- Held regular office hours for Advanced Electrodynamics and Quantum Mechanics courses
- Graded homework and exams
- Created solutions and grading rubrics for homework assignments
Teaching Assistant, University of Utah Aug - Dec 2023
- Lead recitations for Intermediate Electrostatics and Quantum Mechanics course
- Held regular office hours
- Graded homework and exams
- Created solutions and grading rubrics for homework assignments
Teaching Assistant, University of Utah Jan - Apr 2023
- Lead recitations for Modern Physics course
- Held regular office hours
- Graded homework and exams
- Created solutions and grading rubrics for homework assignments
Teaching Assistant, University of Utah Aug - Dec 2022
- Lead recitations for Algebra based Physics I class
- Held regular office hours
STEM Teacher, Wy'East Mountain Academy, Sandy, OR Aug 2021- May 2022
- Taught STEM classes including Physics, Precalculus, Algebra, and Geometry
Long Term Substitute Physics Teacher, Hadley Public Schools, Hadley, MA Oct - Dec 2020
- Taught Introductory Physics, AP Physics I, and Geology
- Created lesson plans, laboratory experiments, homework, and classwork assignments

Mentoring

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| Taona Maphosa , REU Student, Auburn University | May 2026 - Aug 2026 |
| Isabella Carlton , Graduate Student, University of Utah | Aug 2025 - Dec 2025 |
| • Tutoring in quantum mechanics concepts for preparation of the qualifying exam.
Hayley Coluccio , Post-Baccalaureate Student, University of Utah | Aug 2023 - May 2024 |
| • Tutoring in math methods for advanced undergraduate physics coursework.
Chance Hoskinson , Post-Baccalaureate Student, University of Utah | Aug 2023 - May 2024 |
| • Tutoring in math methods for advanced undergraduate physics coursework. | |

Service

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| Peer Reviewer , Journal of Women and Minorities in Science and Engineering | Mar 2026 - Present |
| Community Committee , Department of Physics, Auburn University | Jan 2026 - Present |
| Peer Reviewer , Physical Review Physics Education Research | Apr 2025 - Present |
| Teaching Assistant Committee , Department of Physics & Astronomy, University of Utah | May 2025 - May 2026 |
| • Assigned 52 graduate and undergraduate teaching assistants to roles across 27 courses. | |

Organizations

American Association of Physics Teachers (AAPT)	2024-Present
American Physical Society (APS)	2023-Present
Physics and Astronomy Society for Support and Advocacy for Gender Equity (PASSAGE), University of Utah	2022-2025
Society of Physics Students (SPS), WPI	2016-2019

Outreach

Women in STEM Club Advisor, Wy'East Mountain Academy, Sandy, OR	Sept 2021-May 2022
STEM Started Academy Mentor, Mount Wachusett Community College, MA	July 2018
WPI Touch Tomorrow Science Festival, WPI, Worcester, MA	June 2018